

Q1. DIRECTIONS *for questions 1 and 2:* Type in your answer in the input box provided below the question.

A quadratic function $f(x)$ attains its minimum value of -15 at $x = 3$. If $f(0) = 5$, find the value of $f(9)$.

Q2. DIRECTIONS *for questions 1 and 2:* Type in your answer in the input box provided below the question.

If $2^{171} + n$ is divisible by 105, then what is the least positive integer value of n ?

Q3. DIRECTIONS *for questions 3 to 8:* Select the correct alternative from the given choices.

If $a + b + c + d = 13$, find the minimum possible value of $a^3 + b^3 + c^3 + d^3$, given that a, b, c and d are all positive integers.

☐ a) 142

☐ b) **145**

☐ c) 108

☐ d) **127**

Q4. DIRECTIONS for questions 3 to 8: Select the correct alternative from the given choices.

If P_1 , P_2 and P_3 are the lengths of the altitudes of a triangle ABC, then how many of the following ratios of P_1 , P_2 and P_3 are possible?

i. $P_1 : P_2 : P_3 = 1 : 3 : 3$

ii. $P_1 : P_2 : P_3 = 2 : 4 : 5$

iii. $P_1 : P_2 : P_3 = 3 : 4 : 8$

iv. $P_1 : P_2 : P_3 = 3 : 5 : 8$

☐ a) 2

☐ b) 3

☐ c) 0

☐ d) 1

Q5. DIRECTIONS *for questions 3 to 8:* Select the correct alternative from the given choices.

The ratio of the ages of A and B twelve years ago was 2 : 3 and the ratio of the ages of B and C 12 years from now will also be 2 : 3. If the average age of A and C at present is 57 years, the ratio of the ages of B and C was/will be 1 : 2

- ☐ a) 6 years ago.
- ☐ b) 12 years ago.
- ☒ c) 18 years ago. ✓ **Your answer is correct**
- ☐ d) 18 years from now.

Q6. DIRECTIONS *for questions 3 to 8:* Select the correct alternative from the given choices.

A sequence of terms, $a_1, a_2, a_3, \dots, a_i$, is generated as follows. The first term (a_1) is chosen as 1. To obtain every subsequent term a coin is tossed. If a heads (H) appears, the next term is obtained by adding 2 to the previous term. If a tails (T) appears, only 1 is added to the previous term to obtain the next term. For example, the sequence corresponding to HTH is 1, 3, 4, 6. If $a_{11} = 15$, what is the maximum number of consecutive tails that could have appeared up to that term?

- ☐ a) 6
- ☐ b) 9
- ☐ c) 7
- ☐ d) None of the above

Q7. DIRECTIONS for questions 3 to 8: Select the correct alternative from the given choices.

If $A = \{x : x \leq 10 \text{ and } x \in \mathbb{N}\}$, $B = \{2\}$ and $C = \{7\}$ and S is a subset of A such that $n(S \cap B) = 1$ and $n(S \cap C) = 0$, how many such subsets are possible?

☐ a) 1023

☐ b) **255**

☐ c) 818

☐ d) 256

Q8. DIRECTIONS *for questions 3 to 8:* Select the correct alternative from the given choices.

In a city, the local shuttle trains start from and arrive at the station at fixed intervals and run at a uniform speed. A boy was walking down the railway track at a certain speed. Every half an hour, a local train overtook him and every 20 minutes, a local train passed him in the opposite direction. Find the time interval between a local train passing a certain point on the railway track and the immediately next local train passing that point in the same direction.

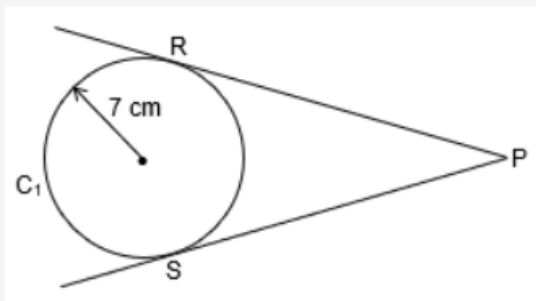
- ☐ a) 26 minutes
- ☐ b) **24 minutes**
- ☐ c) 27 minutes
- ☐ d) 25 minutes

Q9. DIRECTIONS *for question 9:* Type in your answer in the input box provided below the question.

The number of ordered triplets (x, y, z) , such that x, y, z are distinct prime numbers and $xy + yz + zx = 120$ is

Q10. DIRECTIONS for questions 10 to 12: Select the correct alternative from the given choices.

In the figure given below, PR and PS are tangents drawn to circle C_1 , whose radius is equal to 7 cm. If the length of PS is 24 cm, find the area (in sq. cm) of the triangle PRS.



- ☐ a) $\frac{96768}{625}$
- ☐ b) $\frac{93524}{625}$
- ☐ c) $\frac{98756}{625}$
- ☐ d) $\frac{97867}{625}$

Q11. DIRECTIONS *for questions 10 to 12:* Select the correct alternative from the given choices.

In a number system to the base 20, the letters A, B, C, to J represent the values 10, 11, 12, to 19 (to the base 10) respectively. Find the value of $[(\text{HEAD})_{20} - (\text{DEAD})_{20}]$.

- ☐ a) $(\text{D000})_{20}$
- ☐ b) $(32000)_{20}$
- ☐ c) $(32000)_{10}$
- ☐ d) $(4000)_{10}$

Q12. DIRECTIONS *for questions 10 to 12:* Select the correct alternative from the given choices.

In a nuclear reaction, two species A and B are formed at different times. A reference time $t_0 = 0$ is selected. At time t equal to $-2 \mu\text{s}$ (microseconds), the age of A is $18 \mu\text{s}$ more than three times that of B. When is the age of A exactly three times that of B?

☐ a) $t = -7 \mu\text{s}$

☐ b) $t = 9 \mu\text{s}$

☐ c) $t = -9 \mu\text{s}$

☐ d) $t = 7 \mu\text{s}$

Q13. DIRECTIONS *for questions 13 and 14:* Type in your answer in the input box provided below the question.

The 342nd term of the series 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, is

Q14. DIRECTIONS *for questions 13 and 14:* Type in your answer in the input box provided below the question.

Let P represent the set of all natural numbers from 1 to 100, each of which differs from the sum of its factors by 1, and let Q represent the set of all natural numbers from 1 to 100, each of which is equal to the number of its factors. The L.C.M. of the greatest element of P and the greatest element of Q is

Q15. DIRECTIONS *for question 15:* Select the correct alternative from the given choices.

A and B can finish a work, working on alternate days, in 17 days, where A works on the first day. Similarly, they can finish the work, working on alternate days, in $17\frac{2}{3}$ days, where B works on the first day. If C, working alone, can complete the work in 35 days, in how many days can the work be completed when A, B and C work together?

- ☐ a) 6 days
- ☐ b) $6\frac{2}{5}$ days
- ☐ c) $7\frac{2}{5}$ days
- ☐ d) 7 days

Q16. DIRECTIONS *for question 16:* Type in your answer in the input box provided below the question.

Find the area (in sq. units) of the region bounded by the graph $|x - 7| + |y - 9| = 12$.

Q17. DIRECTIONS *for question 17:* Select the correct alternative from the given choices.

L_1 , L_2 and L_3 are three solutions of alcohol and water, with the alcohol percentages (by volume) in them being 20%, $c\%$ and $2c\%$, respectively. If the alcohol percentage (by volume) in the mixture obtained by mixing 1 litre, 3 litres and 2 litres of L_1 , L_2 and L_3 , respectively, is 22%, find the value of c .

☐ a) $12\frac{1}{2}$

☐ b) **16**

☐ c) **20**

☐ d) $16\frac{2}{3}$

Q18. DIRECTIONS for questions 18 and 19: Type in your answer in the input box provided below the question.

$$\text{If } S = \sqrt{1 + \frac{1}{4} + \frac{2}{2}} + \sqrt{1 + \frac{1}{36} + \frac{2}{6}} + \sqrt{1 + \frac{1}{144} + \frac{2}{12}} + \dots + \sqrt{1 + \frac{1}{422500} + \frac{2}{650}}, \text{ find the value of } 26S.$$

Q19. DIRECTIONS *for questions 18 and 19:* Type in your answer in the input box provided below the question.

If the sum of six integers x_1, x_2, x_3, x_4, x_5 and x_6 is 9753 and $E = \sum_{i=1}^6 (-1)^{x_i}$, what is the least possible value of E ?

Q20. DIRECTIONS *for questions 20 to 22:* Select the correct alternative from the given choices.

In a geometric progression, the sum of the first n terms is 3069 and the first term is 3. If the n^{th} term in the progression is 1536, find the common ratio.

☐ a) $\frac{1}{2}$

☐ b) $\frac{3}{2}$

☐ c) **2**

☐ d) **4**

Q21. DIRECTIONS *for questions 20 to 22:* Select the correct alternative from the given choices.

Two classmates P and Q have a conversation as below.

P: $\frac{7}{11}^{\text{th}}$ of my classmates are females.

Q: That is interesting, because $\frac{12}{19}^{\text{th}}$ of my classmates are females.

Assuming that both P and Q speak the truth, which of the following statements are true?

I. Q is a female.

II. P is a female.

III. There are a total of 209 students in the class.

☐ a) Only II and III

☐ b) **Only I**

☐ c) **Only I and III**

☐ d) **None of the above**

Q22. DIRECTIONS *for questions 20 to 22:* Select the correct alternative from the given choices.

If a copper wire of diameter 1.5 cm and length 80 m is melted completely and recast to form a solid spherical ball of radius R cm, then find the value of R (in cm).

☒ a) 15 ✓ **Your answer is correct**

☐ b) 12

☐ c) 20

☐ d) 25